



# United International University (UIU)

Dept. of Computer Science & Engineering (CSE)

CSE 1110: Introduction to Computer Systems Fall 2024

Exam Name: Final Exam Time: 50 Minutes Total Marks: 25

Set B

*Any examinee found engaging in unfair practices will be expelled from the trimester / program as per UIU disciplinary rules.*

Name:

ID:

Note: **Answer all the questions.** The numbers on the right side indicate marks for the respective question.

| 1.                                 | <p>You have to take four integers (<b>a, b, p, q</b>) as input and find the solutions to the equation <math>ax^2 + bx + c = 0</math>. The value of c can be determined by taking the remainder when p is divided by q. You can safely assume that the solutions to the given equation will be real. To find the solution, you can use the quadratic formula.</p> <p><b>Quadratic Formula:</b> The solutions to the equations <math>ax^2 + bx + c = 0</math> are <math>x_1</math> and <math>x_2</math> where <math>x_1 = (-b + \sqrt{D})/2a</math>, <math>x_2 = (-b - \sqrt{D})/2a</math> and <math>D = b^2 - 4ac</math>.</p> <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>Enter four integers:<br/>2 -8 15 4</td><td>3.581 0.419</td></tr><tr><td>Enter four integers:<br/>5 -10 12 5</td><td>1.775 0.225</td></tr></table> | Sample Input | Sample Output | Enter four integers:<br>2 -8 15 4 | 3.581 0.419 | Enter four integers:<br>5 -10 12 5 | 1.775 0.225 | [CO4] 9 |     |      |         |         |
|------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|---------------|-----------------------------------|-------------|------------------------------------|-------------|---------|-----|------|---------|---------|
| Sample Input                       | Sample Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |               |                                   |             |                                    |             |         |     |      |         |         |
| Enter four integers:<br>2 -8 15 4  | 3.581 0.419                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |               |                                   |             |                                    |             |         |     |      |         |         |
| Enter four integers:<br>5 -10 12 5 | 1.775 0.225                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |              |               |                                   |             |                                    |             |         |     |      |         |         |
| 2.                                 | <p>Write a C program that takes a character and an integer number as input from the user. If the character is ‘E’ or ‘e’ and the number is even, print “Even”, or “Invalid” otherwise. If the character is ‘O’ or ‘o’ and the number is odd, then print “Odd”, or “Invalid” otherwise. Solve <b>using only if/else</b>. You can safely assume that the character input is valid.</p> <table><tr><th>Sample Input</th><th>Sample Output</th></tr><tr><td>E 20</td><td>Even</td></tr><tr><td>e 33</td><td>Invalid</td></tr><tr><td>O 65</td><td>Odd</td></tr><tr><td>o 34</td><td>Invalid</td></tr></table>                                                                                                                                                                                                                                         | Sample Input | Sample Output | E 20                              | Even        | e 33                               | Invalid     | O 65    | Odd | o 34 | Invalid | [CO4] 8 |
| Sample Input                       | Sample Output                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |              |               |                                   |             |                                    |             |         |     |      |         |         |
| E 20                               | Even                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |              |               |                                   |             |                                    |             |         |     |      |         |         |
| e 33                               | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |               |                                   |             |                                    |             |         |     |      |         |         |
| O 65                               | Odd                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |              |               |                                   |             |                                    |             |         |     |      |         |         |
| o 34                               | Invalid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |              |               |                                   |             |                                    |             |         |     |      |         |         |

3.

You are a **cashier at a bookstore** where the store offers **three books**, each with a **fixed price**:

- 1. English (280 Taka)**
- 2. Math (310 Taka), and**
- 3. ICT (300 Taka).**

**As a CSE student, you want to automate the price calculation.** Ask the user to select a book by entering a number. The user must **select a book by entering a number** corresponding to their choice and then enter the **quantity (integer)** they wish to purchase. The system calculates the **total price** and adds a **10% tax** to the amount. If the quantity is greater than **3**, apply **7% discount**. Finally, the program displays the **tax amount, discount amount (if any), and the final payable price**.

Write a C program with the above features **using only switch case**.

| Sample Input                                                                                         | Sample Output                                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ----- Welcome to Book Store -----<br><br>Enter your selected book choice: 3<br>Enter the quantity: 2 | You selected ICT book, which price is 300 taka<br><br>Tax Amount = 60 taka<br>Discount Amount = 0 taka<br>2 Books total price is 660 taka<br><br>----- Thank you for being with us ----- |

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